

BiasAperture — Weekly Project Report (WK4)

Period: 2026-08-22 through 2026-08-27 (week ending Thursday) **Project:** BiasAperture — A Diagnostic and Evaluative Framework for Auditing Demographic Bias in Facial Analysis Systems
Program: Fusemachines AI Fellowship (AIF) 2026 (Kathmandu, Nepal)
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Status: M1–M4 Core Engine Verified (55/55 Tests Passing) · M5 Integration in Progress · Multi-Remote Clean

1. Executive Summary & Objectives

Week 4 represents the primary implementation, empirical validation, and audit milestone of the BiasAperture project. Over this period, the team executed a 20-track parallel research sprint, established an auditable claim ledger, performed empirical raw-data exploration of the FairFace and UTKFace benchmarks, implemented the complete fairness detection engine (Fairlearn & native AIF360 backends), implemented BCa bootstrap confidence intervals and χ^2 significance tests, developed the additive linear Shapley feature attribution layer, completed the offline Jinja2 report generator and CLI orchestrator, reached 55/55 passing unit tests, and conducted an end-to-end repository audit synchronized across all Git remotes.

Key Objectives

- Execute the 20-track parallel research sprint and synthesize findings into 3-level architecture specifications (High, Mid, Low level).
- Establish the auditable `CLAIM_LEDGER.md` with strict verification lifecycle and author comprehensive Viva defense guides.
- Conduct empirical dataset profiling on FairFace and UTKFace to identify disk counts, labeling anomalies, and DEX synthetic noise.
- Implement the core fairness metrics engine, Fairlearn + native AIF360 backend harmonization, BCa bootstrap ($B \geq 1,000$ $\geq 1,000$), and χ^2 significance tests ($p < 0.05$ $p < 0.05$).
- Implement the additive Shapley surrogate attribution layer ($\phi_i = w_i(x_i - \mathbb{E}[x_i])$) in `explainability.py` with robust Windows DLL/MAX_PATH guards.
- Connect all components into the `bias-aperture` CLI orchestrator and standalone offline HTML report generator.
- Achieve 100% passing test coverage (55 tests) and resolve all merge conflicts across upstream Git remotes (`origin` , `duo` , `org`).

2. Work Packages & Milestone Mapping

Milestone / Workstream	Focus & Scope	Primary Owner	Status
20-Track Research Sprint	Tracks 01–20 execution, 3-level synthesis reports, and architecture PDFs	Aaradhya	Completed
Claim Ledger & Defense Guides	CLAIM_LEDGER.md , PROPOSAL_DEFENSE_GUIDE.md , PRE_PROPOSAL_READING_GUIDE.md	Aaradhya	Completed
Data Exploration & Profiling	explore_fairface.py , explore_utmface.py , data_exploration_report.md (97,698 FairFace images verified)	Tisha	Completed
WP4: Core Fairness Engine	DPD, DIR, EOP, EOD, Fairlearn adapter, and native AIF360 BinaryLabelDataset / ClassificationMetric	Aaradhya	Completed
WP4: Statistical Rigor	95% BCa bootstrap confidence intervals ($B \geq 1,000$) and χ^2 significance testing	Aaradhya	Completed
WP4 (FR-005): Explainability	Exact linear Shapley surrogate explainer (ShapExplainerEngine) with surrogate fallback	Aaradhya	Completed
WP3: Offline HTML Report	Standalone zero-network Jinja2 HTML report generator with embedded Base64 charts	Aaradhya	Completed
WP5: CLI & Orchestration	bias-aperture audit CLI with --explain , --backend , and --bca-bootstrap flags	Aaradhya	Completed
Verification & Quality	55/55 Pytest unit test suite, Ruff linting & formatting (0 errors)	Joint	Completed
Multi-Remote Git Sync	Resolving upstream divergence and syncing across origin , duo , and org	Joint	Completed

3. Detailed Accomplishments & Technical Highlights

3.1 20-Track Parallel Research Sprint & 3-Level Syntheses

- Executed 20 targeted research investigations covering demographic taxonomies, intersectional binning, metric trade-offs, statistical power, bootstrap validity, surrogate modeling, and regulatory compliance.
- Synthesized findings into three authoritative specification documents:
 - HIGH_LEVEL_SYNTHESIS.md : Executive summary and statutory mapping.
 - MID_LEVEL_ARCHITECTURE.md : Component interactions, data flow, and interface contracts.
 - LOW_LEVEL_SPECIFICATION.md : Mathematical equations, algorithmic bounds, and error handling.

- Formalized the auditable `CLAIM_LEDGER.md` tracking 60+ empirical assertions through ASSERTED →\to VERIFIED →\to REPRODUCIBLE states.

3.2 Dataset Profiling & Empirical Discoveries

- Authored exploration scripts `scripts/explore_fairface.py` and `scripts/explore_utkface.py`.
- Uncovered that FairFace contains **97,698 released images** on disk across train/validation splits, correcting paper citations of 108.5k.
- Discovered systematic DEX age label noise and label boundary artifacts in UTKFace, providing concrete empirical justification for designating UTKFace as a secondary validation testbed rather than primary benchmark.

3.3 Native AIF360 Backend Harmonization (`src/bias_aperture/fairness/backends.py`)

- Replaced adapter pass-through with native `aif360.datasets.BinaryLabelDataset` and `aif360.metrics.ClassificationMetric`.
- Successfully harmonized metric definitions between Fairlearn and AIF360 to ensure consistent cross-backend validation.

3.4 Exact Additive Shapley Surrogate Explainability

(`src/bias_aperture/explainability.py`)

- Implemented linear surrogate feature attribution computing exact Shapley values:

$$\phi_i = w_i(x_i - \mathbb{E}[x_i])$$
- Built defensive import and execution wrappers safeguarding against Windows C-extension DLL and MAX_PATH loading failures.
- Added automatic attribution triggers on statistically significant disparities ($p < 0.05, n \geq 30$ or $p < 0.05, n \geq 30$).

3.5 Test Suite & Code Quality (55/55 Tests Passing)

- Expanded the automated test suite to 55 comprehensive tests:

```
src/tests/test_backend_harmonization.py .... [ 7%]
src/tests/test_cli.py ..... [ 18%]
src/tests/test_data_ingestion.py .... [ 25%]
src/tests/test_explainability.py ... [ 30%]
src/tests/test_fairness_backends.py ..... [ 41%]
src/tests/test_fairness_metrics.py ..... [ 60%]
src/tests/test_fairness_statistics.py ..... [ 76%]
src/tests/test_known_answer_fairness_metrics.py .... [ 83%]
src/tests/test_model_interface.py ... [ 89%]
src/tests/test_offline_report_contract.py ... [ 94%]
src/tests/test_report_generator.py . [ 96%]
src/tests/test_schema.py .. [100%]
===== 55 passed in 3.76s =====
```

- Ruff linter and formatter: 0 errors across 27 source and test files.

4. Master TO DOs & Remaining Roadmap

1. Benchmark Inference Run (M5):

- Run ResNet-34 inference over FairFace validation split (10,954 images) to generate `data/processed/fairface_predictions_val.csv`.

2. Flagship Audit HTML Generation:

- Execute `bias-aperture audit` on live predictions to generate full `report/audit_report.html`.

3. **Supervisor Logistics & Marking Scheme:**

- Confirm capstone defense date, presentation duration, and grading criteria with TA Shreejan Kisee.

4. **LaTeX Report Integration:**

- Insert final computed disparity tables, BCa confidence intervals, and SHAP attribution plots into `report/main.pdf`.

5. **Viva Defense Rehearsal:**

- Conduct mock defense using `PROPOSAL_DEFENSE_GUIDE.md` and `VERIFICATION_AND_SCRUTINY_GUIDE.md`.